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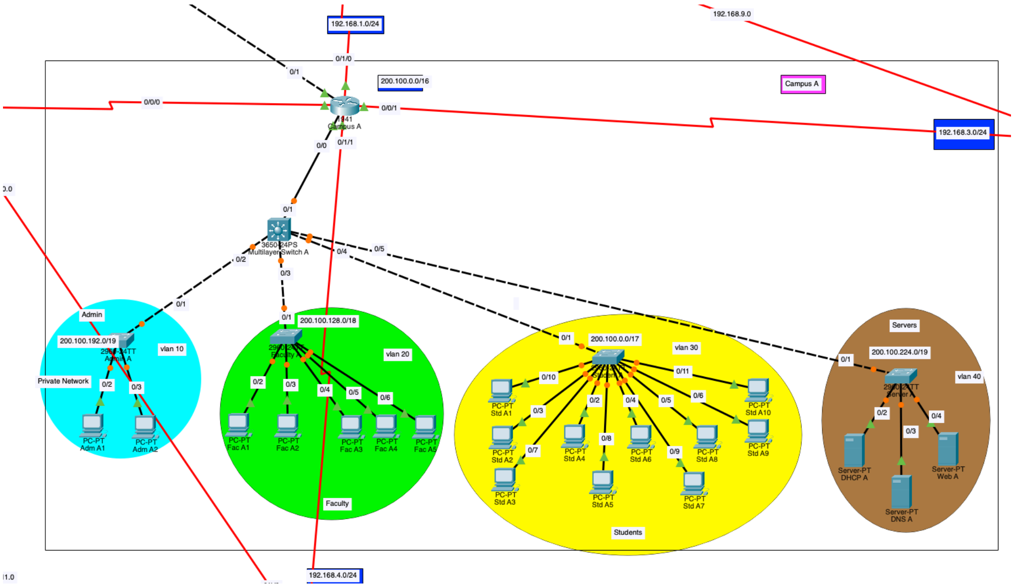
Section	BCS-5F
Subject	Computer Networks

Final Project

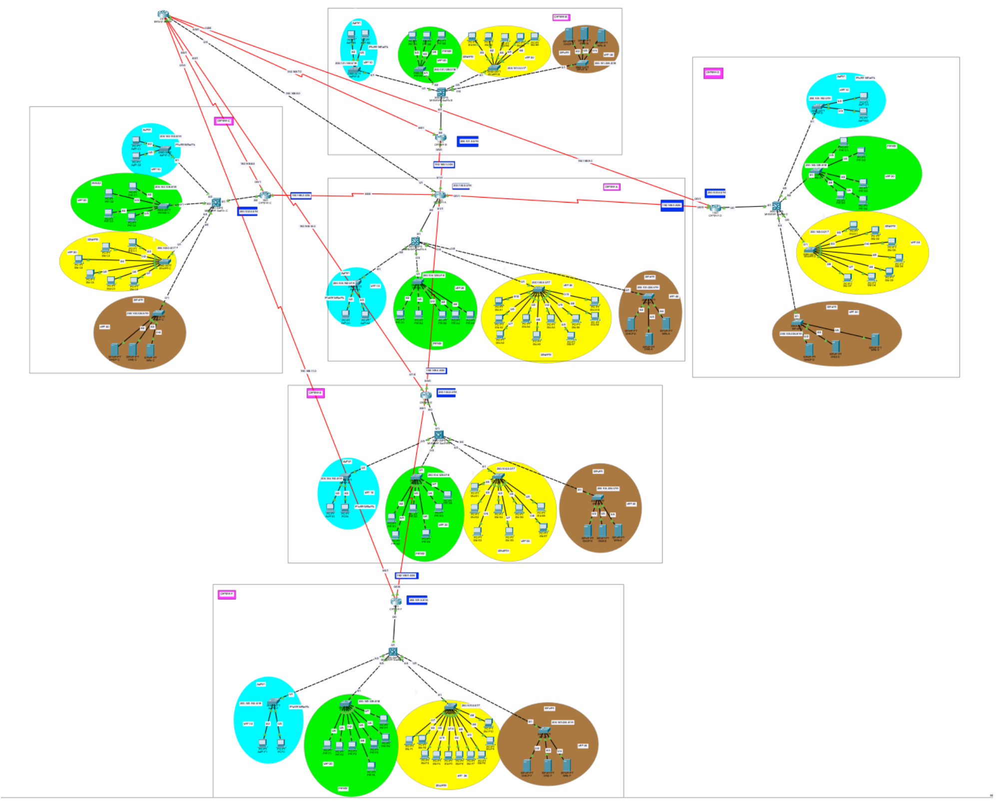
Report

This is Campus A topology.

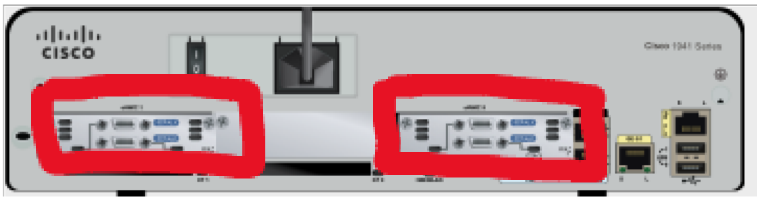
Note: All other 5 campuses topologies are similar.



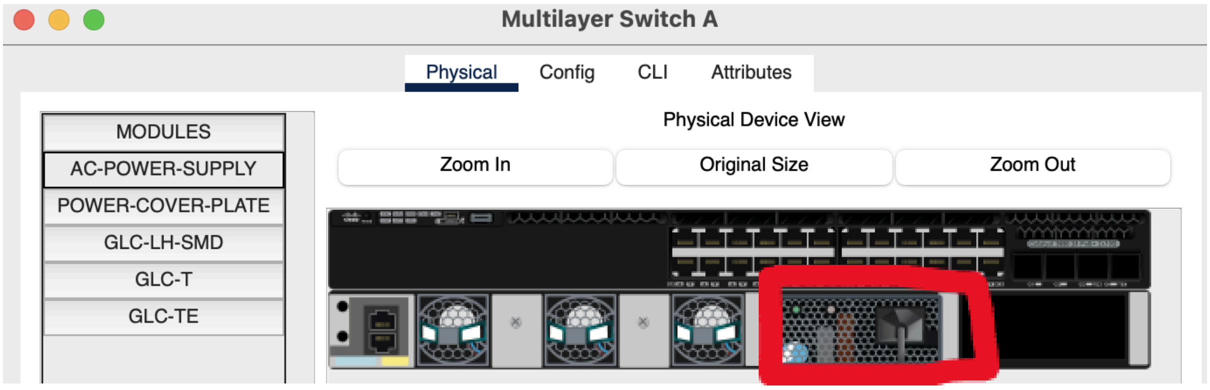
Topology of all 6 campuses.



In router 1941, I had added serial ports to connect all routers to each other.



3650 Multilayer switch to which all department swictches of 1 campus is attached. I had also apply power supply in order of its working.



IP Addressing & Subnetting Table - Including VLSM plans

For Campus A: 200.100.0.0/16, for Campus B: 200.101.0.0/16 and so on for all campuses.

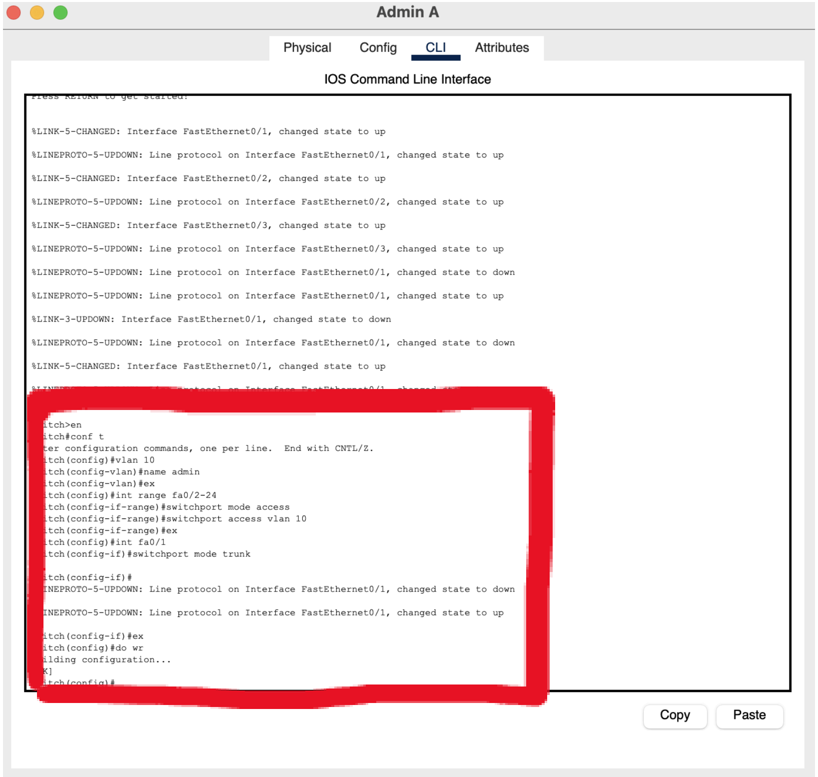
For each department:

Department	Campus A	Campus B	Campus C	Campus D	Campus E	Campus F
Admin	200.100.192.0	200.101.192.0	200.102.192.0	200.103.192.0	200.104.192.0	200.105.192.0
Faculty	200.100.128.0	200.101.128.0	200.102.128.0	200.103.128.0	200.104.128.0	200.105.128.0
Students	200.100.0.0	200.101.0.0	200.102.0.0	200.103.0.0	200.104.0.0	200.105.0.0
Servers	200.100.224.0	200.101.224.0	200.102.224.0	200.103.224.0	200.104.224.0	200.105.224.0

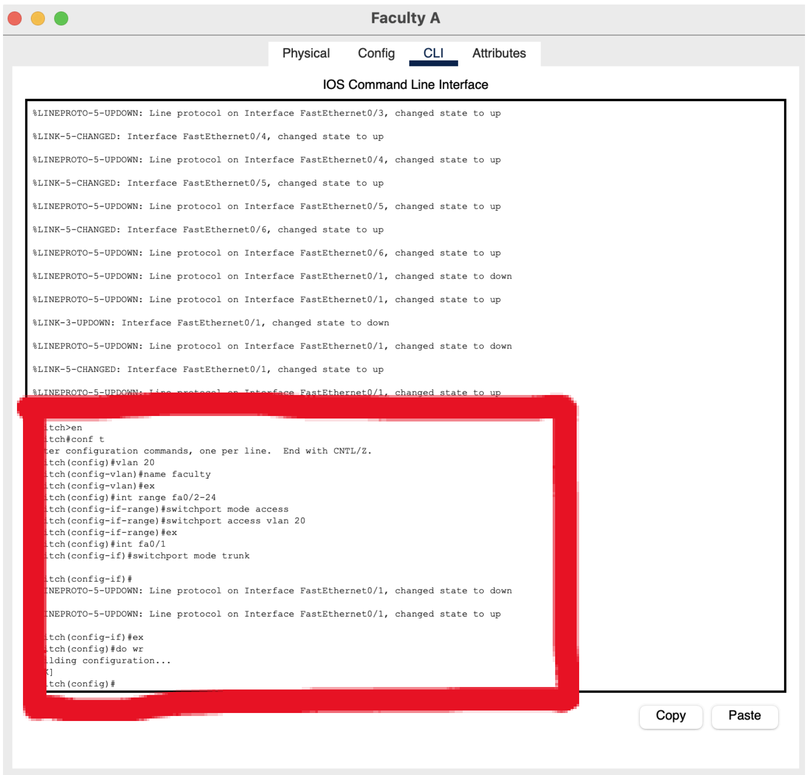
For Campus A to Campus B: 192.168.1.0/24, for Campus A to Campus C: 192.168.2.0 and so in the same for all other campuses for inter communication.

Also all connection of campuses to Backup server uses the same technique of IP allocation like for Backup Server and Campus A: 192.168.6.0/24 and so on for all other campuses.

Vlan for Admin: (Campus A)



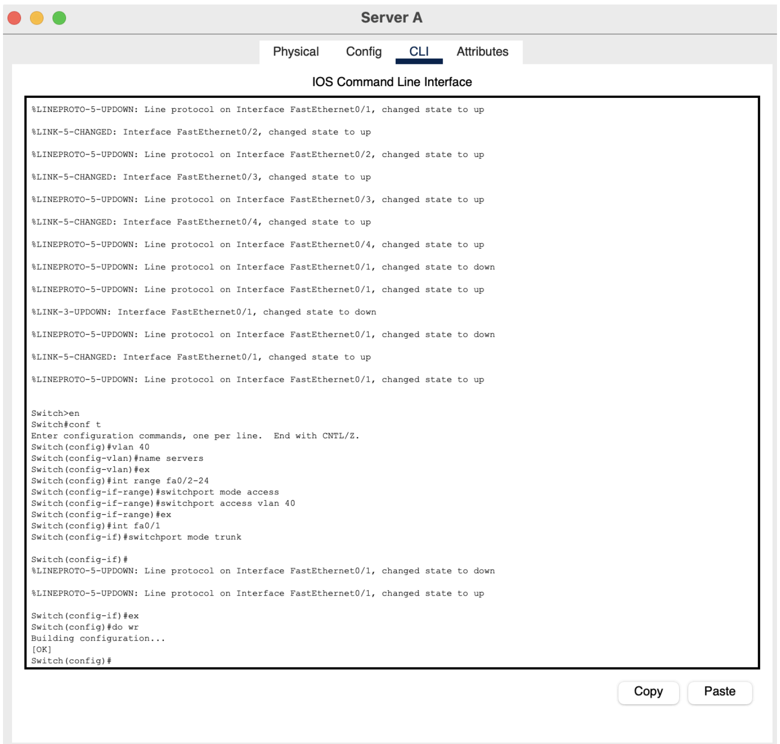
Vlan for Faculty: (Campus A)



Vlan for Students: (Campus A)

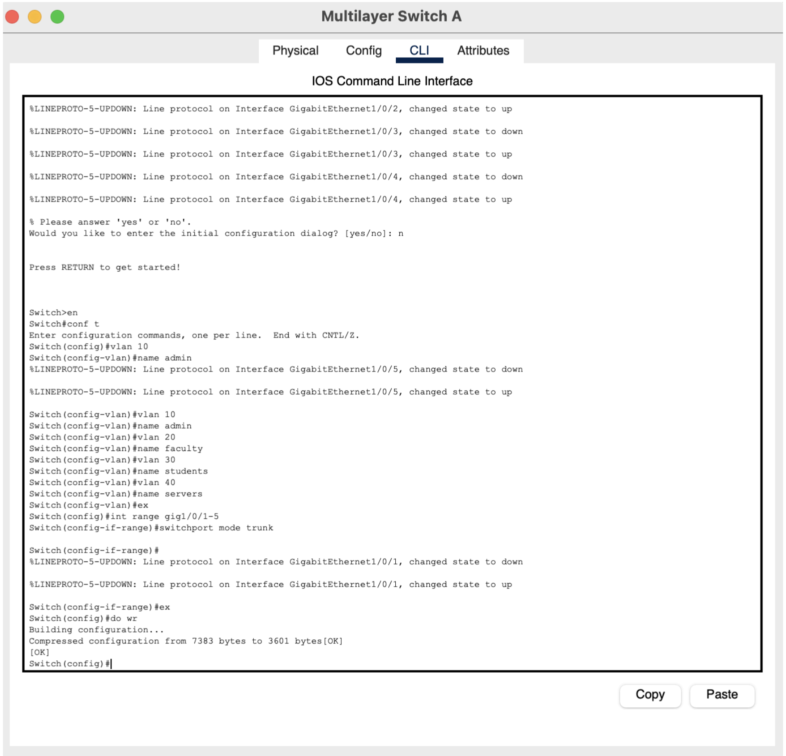

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up


Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#name students
Switch(config-vlan)#ex
Switch(config)#int range fa0/2-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#ex
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#ex
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
```

Vlan for Servers: (Campus A)




```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up


Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 40
Switch(config-vlan)#name servers
Switch(config-vlan)#ex
Switch(config)#int range fa0/2-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#ex
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#ex
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
```

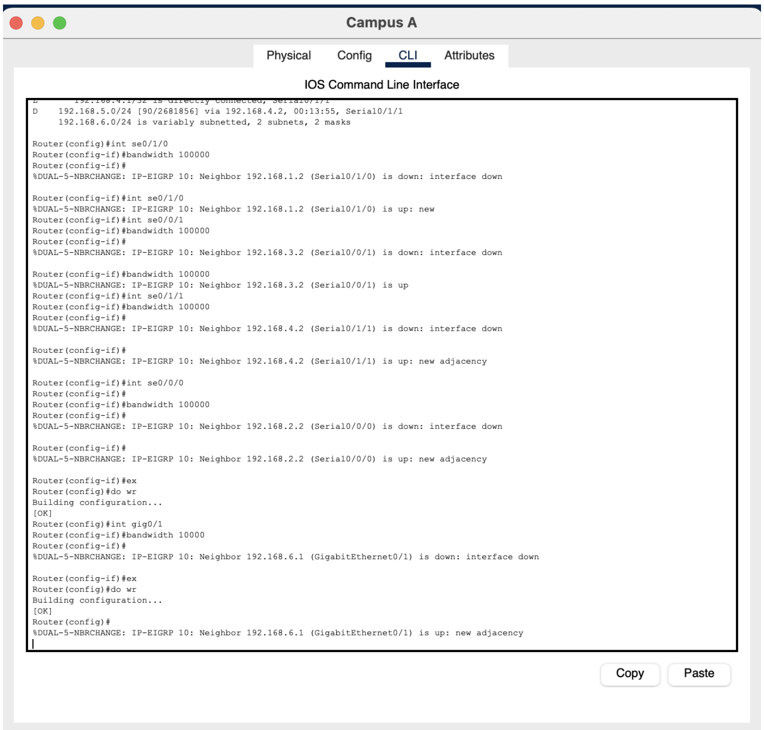
Vlan Configuration at Multilayer Switch:



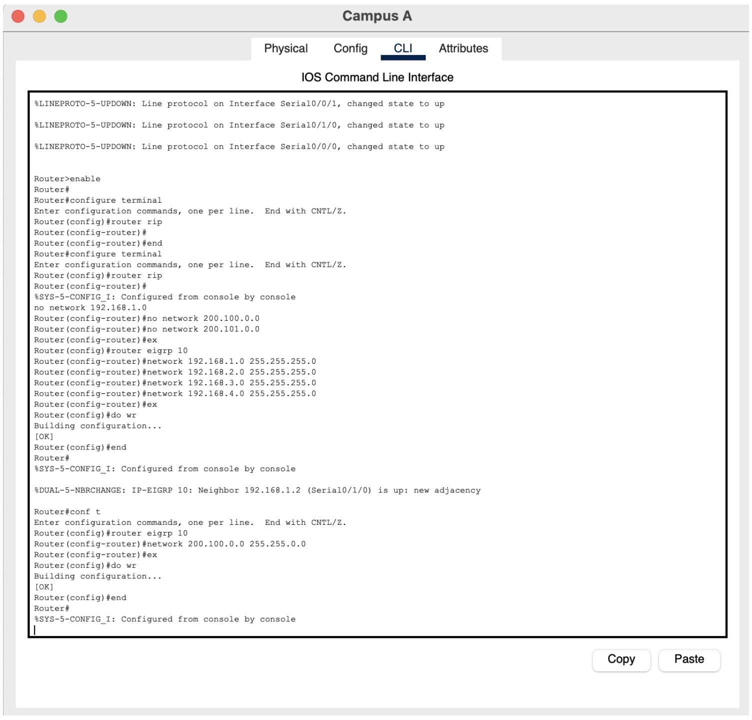
Gateways configuration for each vlan at router:

Note: This (vlan + gateway) is shown for only Campus A. This same way goes for all other campuses.

Set bandwidth of primary higher and backup link slower to make sure eigrp always first ensure primary link.

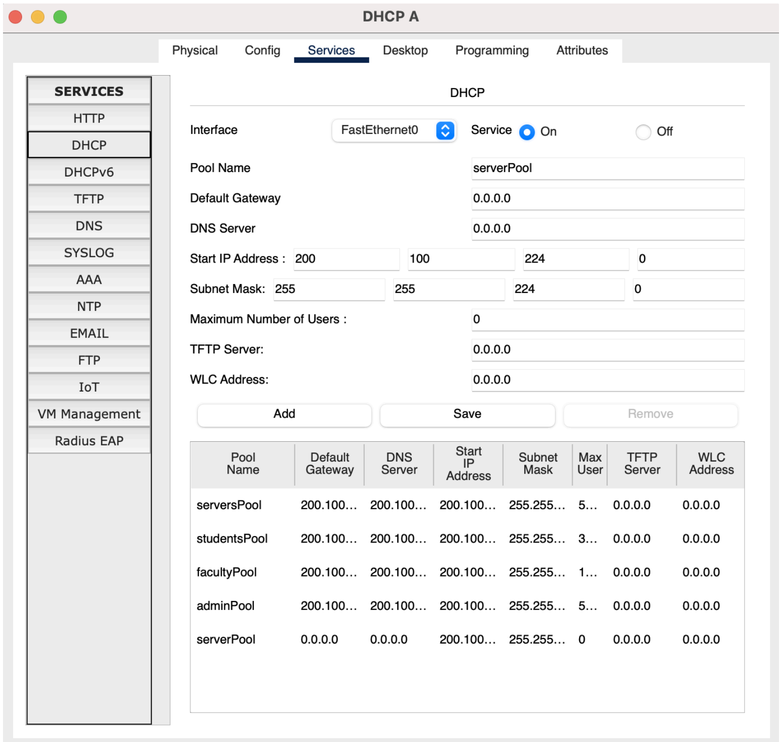


EIGRP at Campus A router:

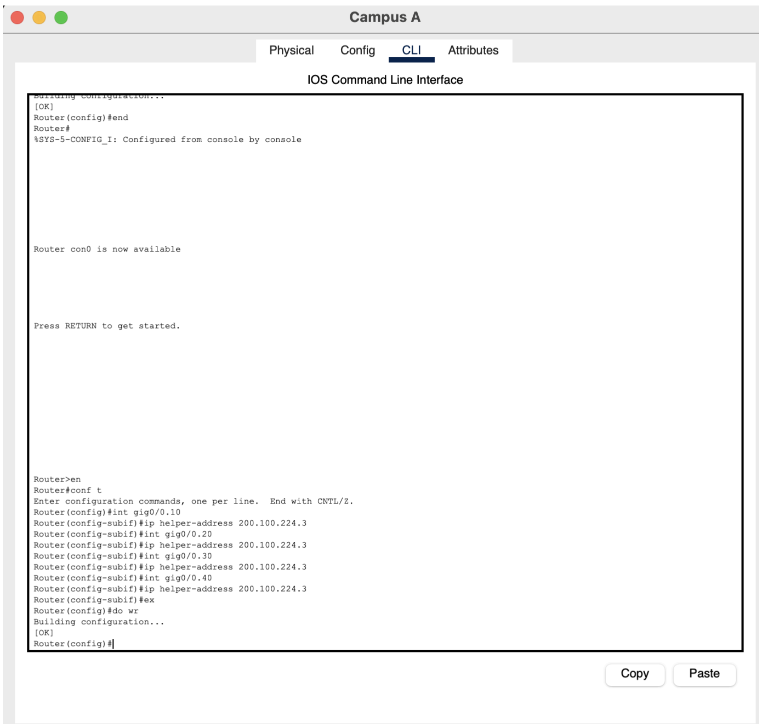


Note: This eigrp is shown only for Campus A. Similary for all other campuses is also implemented.

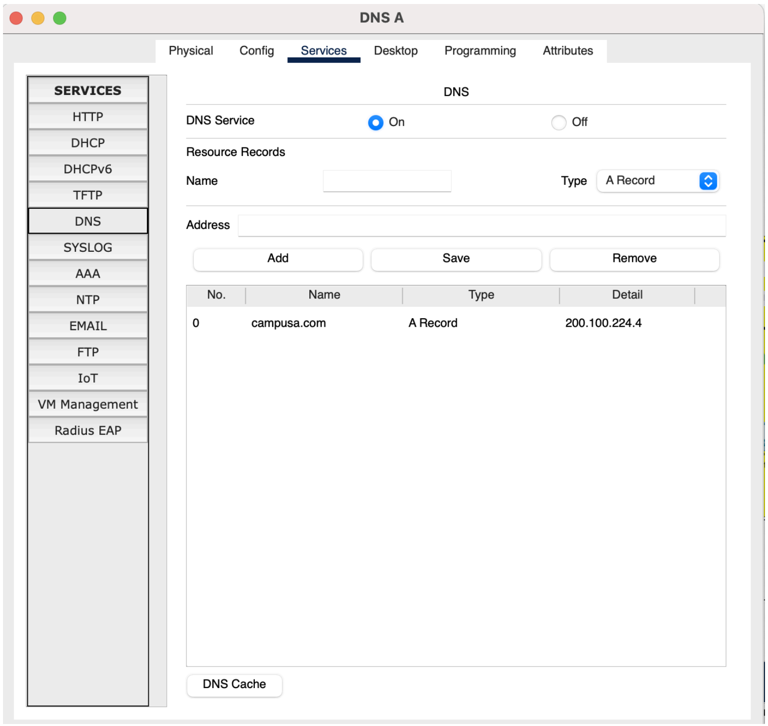
DHCP pools for Campus A:



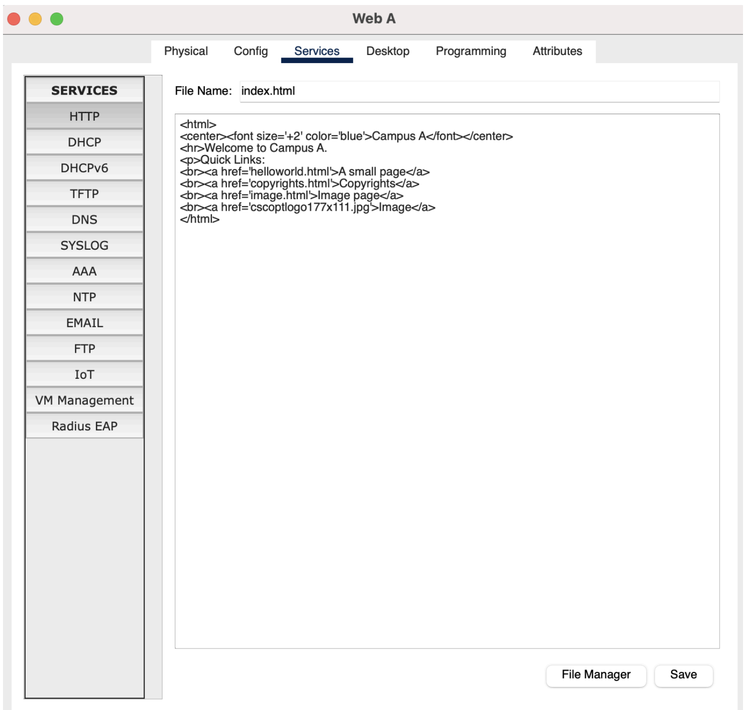
DHCP broadcast message request is set for specifically sending DHCP request to that particular server.



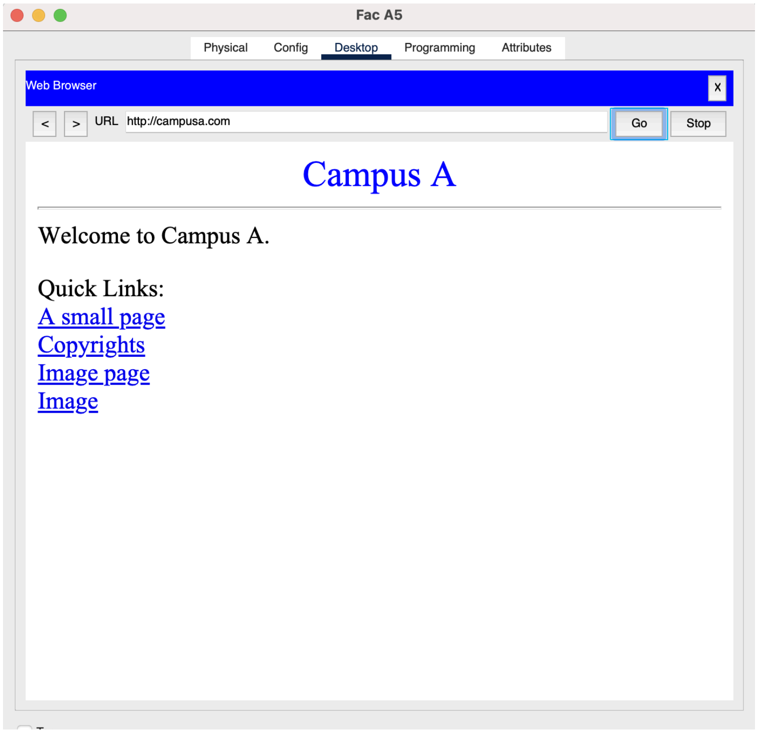
In DNS service, address of Web server is given.



Index.html of Web server is updated for web services of Campus A.



Using web services of Campus A.



Note: All this work is shown only for Campus A. Similar way was adopted for all other campuses.

NAT configuration for Campus A:

Campus A

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0
Router(config-if)#no ip nat inside
Router(config-if)#int se0/0/0
Router(config-if)#no ip nat outside
Router(config-if)#int se0/0/1
Router(config-if)#no ip nat outside
Router(config-if)#int se0/1/1
Router(config-if)#no ip nat outside
Router(config-if)#int se0/1/0
Router(config-if)#no ip nat outside
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip nat inside source static 200.100.192.10 192.168.1.3
Router(config)#ip nat inside source static 200.100.192.11 192.168.1.3
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0.10
Router(config-subif)#ip nat inside
Router(config-subif)#int se0/0/0
Router(config-if)#ip nat outside
Router(config-if)#int se0/0/1
Router(config-if)#ip nat outside
Router(config-if)#int se0/1/1
Router(config-if)#ip nat outside
Router(config-if)#int se0/1/0
Router(config-if)#ip nat outside
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

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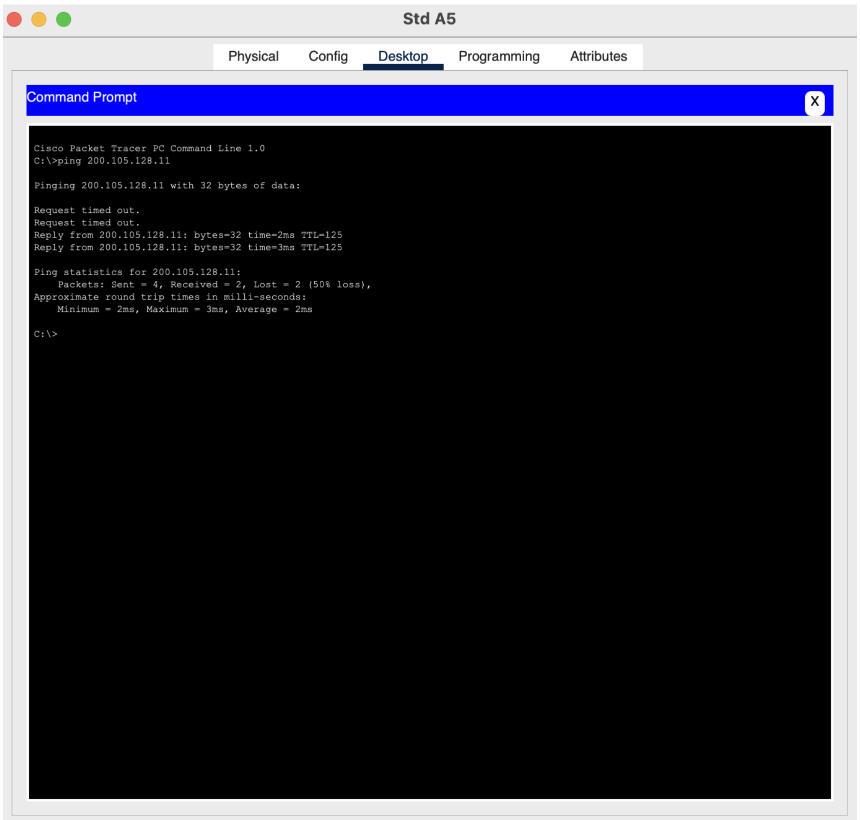
NAT table of Campus A:

NAT Table for Campus A

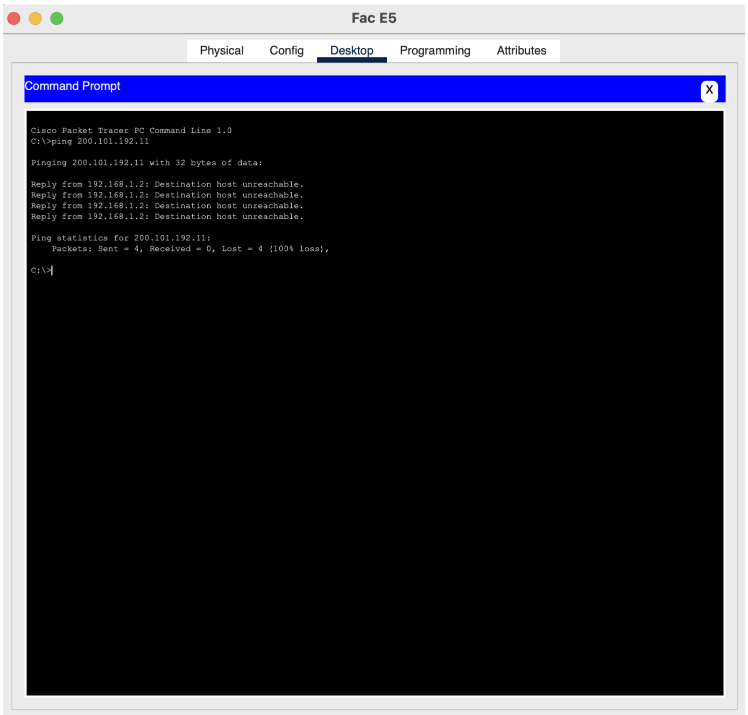
Protocol	Inside Global	Inside Local	Outside Local	Outside Global
---	192.168.1.3	200.100.192.10	---	---
---	192.168.1.4	200.100.192.11	---	---

ACL for NAT of Campus A:

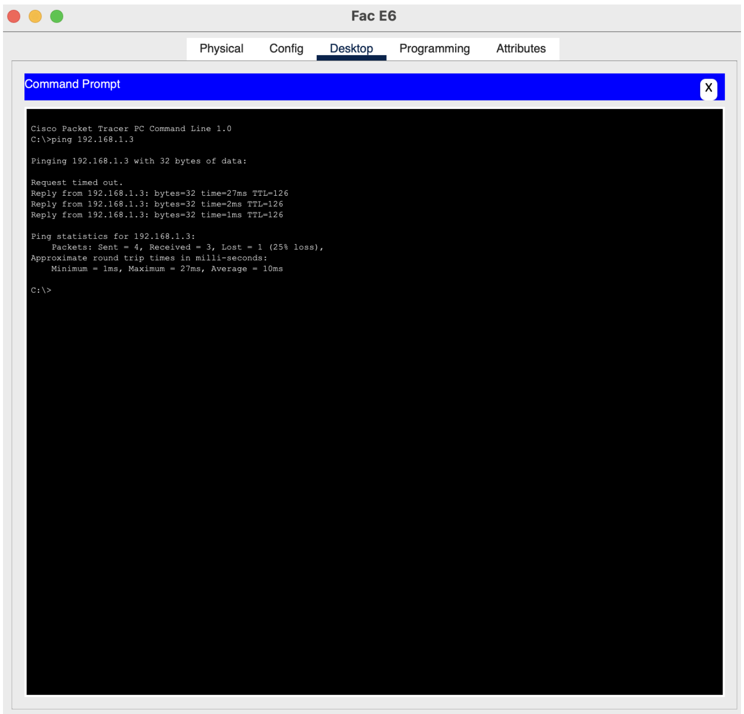
Sending packet from Students of Campus A to Faculty of Campus F:



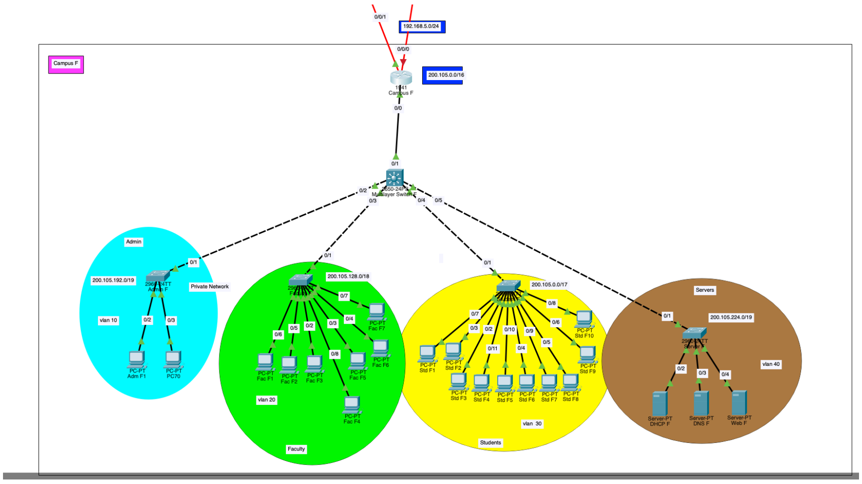
Sending packet from Faculty of Campus E to Admin of Campus B using private address **(Failed)**:



Sending packet from Faculty of Campus E to Admin of Campus B using public address **(Success)**:



I shut down primary link and checked that secondary link working or not?



As the packets are sent, so secondary link is also working.